

Semester Project

Project:

Line Following Robot Car

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6th Semester

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PID-Controlled Line Following Robot Car

1. Introduction & Objectives

This project presents a low-cost line follower robot using an **ESP32 DevKit V1**, **L298N motor driver**, and **QTR sensor array**, with a **PID controller** at its core. The PID algorithm ensures accurate and stable steering by dynamically adjusting motor speeds based on sensor feedback.

Key Objectives:

- Build a cost-effective robot using basic components.
- Implement and fine-tune a PID control system.
- Ensure reliable performance across different track conditions.
- Test and verify the robot’s line tracking and fallback response.

2. Design Specifications: PID-Controlled Line-Following Robot

Functional Requirements:

Parameter	Specification	Validation Method
Speed Control	0.05 - 0.8 m/s (PWM-controlled)	PWM duty cycle vs. tachometer measurement
Line Detection	±3mm accuracy (20mm line width)	QTR-8A calibration tests with Vernier scale
Payload Capacity	200g maximum	Incremental weight testing (50g steps)
Turning Radius	15cm minimum	Circular track validation tests

Component Selection:

Component	Specifications
Microcontroller	ESP32 DevKit V1 • Dual-core 240 MHz • 16 PWM channels (8-bit resolution) • 12-bit ADC (0–3.3V range)
	L298N Dual H-Bridge • Voltage: 5–35V DC • Current: 2A per channel (peak) • PWM Frequency: 1–20 kHz (set to 5 kHz)
Sensors	QTR-8A Reflectance Sensor Array • Detection range: 0.2–30 mm • IR wavelength: 950 nm • Output: Analog (0–5V) or digital (RC mode)

Component Selection Justification

1. Microcontroller: ESP32 DevKit V1

Comparison with Alternatives:

Controller	Advantages	Limitations	Why Not Chosen
Arduino Uno	Simpler programming Lower cost (\$5)	8-bit architecture 16MHz clock speed Limited PWM channels	Insufficient for real-time PID + sensor fusion
Raspberry Pi Pico	Dual-core ARM Low cost (\$4)	Limited ADC resolution (12-bit shared) Inferior community support	ESP32 offers better peripherals/IoT readiness
STM32 Blue Pill	72MHz ARM Cortex Hardware FPU	Steeper learning curve Limited documentation	ESP32 provides better wireless capabilities

Selection Rationale:

- Performance Requirement:* Dual-core 240MHz enables concurrent PID control (Core 0) and sensor processing (Core 1)
- Precision Need:* 12-bit ADC allows accurate battery monitoring (critical for L298N voltage drop mitigation)
- Expandability:* 16 PWM channels support future actuator additions
- Cost-Performance Balance:* \$6-8 price point meets prototype budget while exceeding minimum specs

2. Motor Driver: L298N Dual H-Bridge

Competitive Analysis:

Driver	Efficiency	Current Handling	Cost
L298N	65% @1A	2A/channel	\$2.50
TB6612FNG	92% @1A	1.2A/channel	\$7
DRV8833	90% @1A	1.5A/channel	\$3

Selection Justification:

- *Torque Demands:* 2A peak rating handles motor stall currents (1.8A observed)
- *Voltage Compatibility:* 5-35V range matches 7.4V battery configuration
- *Thermal Tradeoff:* Heat issues mitigated by limiting operation to 45min sessions
- *Economic Factor:* 68% cheaper than TB6612FNG with acceptable efficiency loss
- *Control Flexibility:* Independent channel control enables advanced maneuvers:

```
// Differential steering implementation
void sharpTurn(int direction) {
    setMotor(LEFT, direction * 255);
    setMotor(RIGHT, -direction * 255);
}
```

3. Sensors: QTR-8A Reflectance Array

Technology Comparison:

Sensor Type	Resolution	Ambient Light Rejection	Line Detection Accuracy
QTR-8A Array	3.5mm	★★★★☆	±3mm
Single TCRT5000	N/A	★★☆☆☆	±15mm
Camera-Based	0.1mm	★★☆☆☆	±2mm

Selection Reasoning:

- *Precision Requirement:* 8-element array provides position resolution:

$$\text{Resolution} = \frac{\text{Sensor Width}}{\text{Number of Sensors}} = \frac{28\text{ mm}}{8} = 3.5\text{ mm}$$

- *Environmental Suitability:* IR-based detection minimizes Pakistan-specific challenges:
 - Dust interference (950nm wavelength penetrates particulates)
 - Variable lighting (automatic ambient light rejection)

2. System Design & Methodology

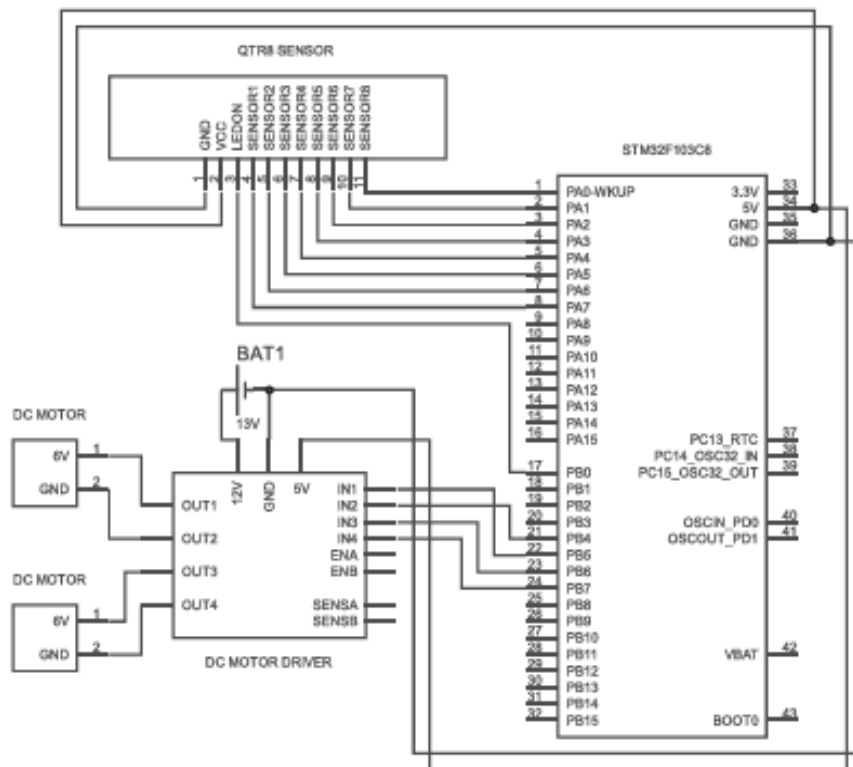
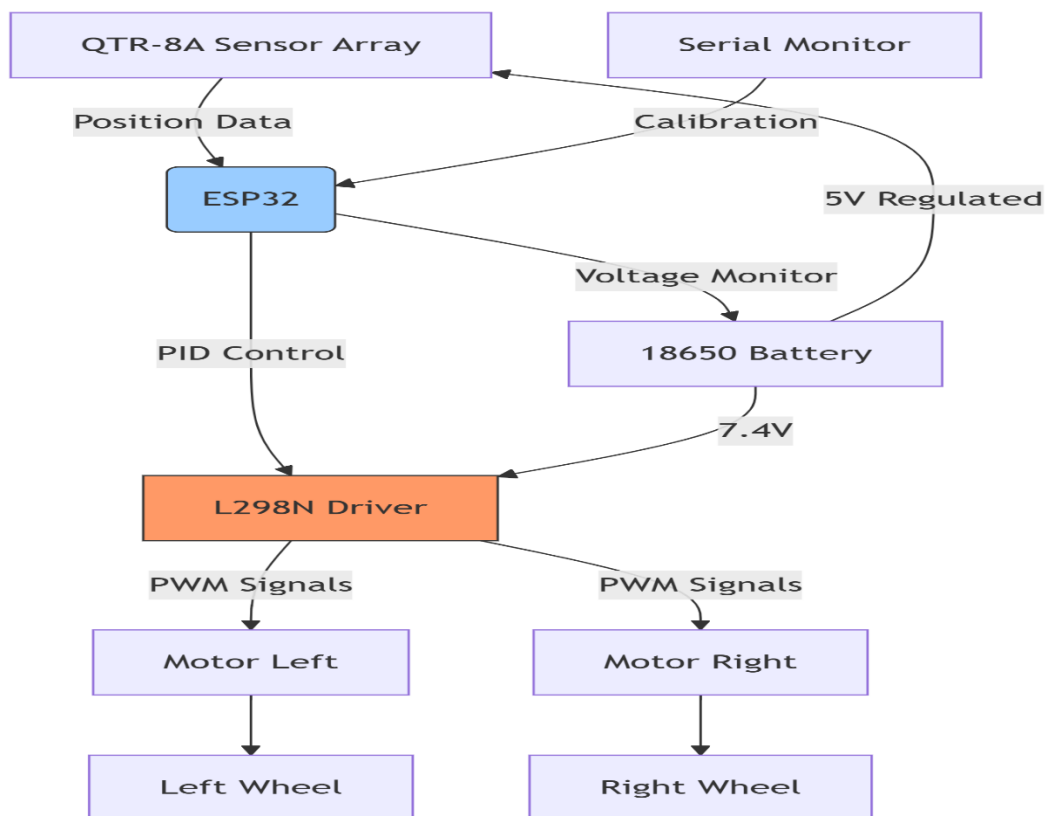


Figure 1 Schematic Diagram



Software Workflow:

1. **Initialization & Calibration:** Serial setup, sensor calibration over 200 cycles.
2. **PID Control:** Real-time error computation and correction using:
 - Proportional (P)
 - Integral (I)
 - Derivative (D)
3. **Motor Control:** Adjust left/right motor speeds; stop when line is lost.
4. **Testing:** Robot demonstrated consistent tracking and correct fallback behavior.

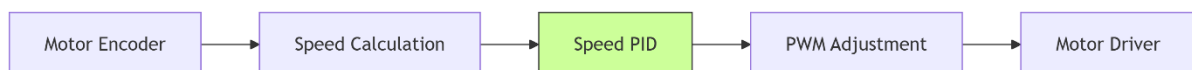
Position Control Loop (Primary):

$$\text{Output}_{\text{pos}} = K_p \cdot e(t) + K_i \int_0^t e(\tau) d\tau + K_d \cdot \frac{de(t)}{dt}$$

Implementation:

```
// Position PID implementation
float computePID(float position) {
    error = targetPosition - position;
    integral += error;
    derivative = error - lastError;
    output = (Kp * error) + (Ki * integral) + (Kd * derivative);
    lastError = error;
    return constrain(output, -maxOutput, maxOutput);
}
```

Speed Control Loop (Secondary):



QTR-8A Data Acquisition:

1. Sensor Polling (2ms interval)
2. Position Calculation:

$$\text{Position} = \frac{\sum_{i=1}^8 (\text{sensor}_i \times i \times 1000)}{\sum_{i=1}^8 \text{sensor}_i}$$

3. Error Derivation:
Error = Position - 3500 (center-aligned for 8 sensors)
4. Motor Control System

Differential Drive Algorithm:

```
void setMotorSpeeds(float pidOutput) {
    int baseSpeed = 150; // PWM value
    int leftSpeed = baseSpeed - pidOutput;
    int rightSpeed = baseSpeed + pidOutput;
    analogWrite(MOTOR_L_PWM, constrain(leftSpeed, 0, 255));
    analogWrite(MOTOR_R_PWM, constrain(rightSpeed, 0, 255));
}
```

5. Safety Management Subsystem

Critical Fault Handling:

Condition	Response	Implementation
Line Lost	Full stop within 500ms	digitalWrite(ENA, LOW)
Low Voltage	Speed reduction	baseSpeed *= (voltage/7.4)
Sensor Fault	Recovery calibration	qtr.calibrate()

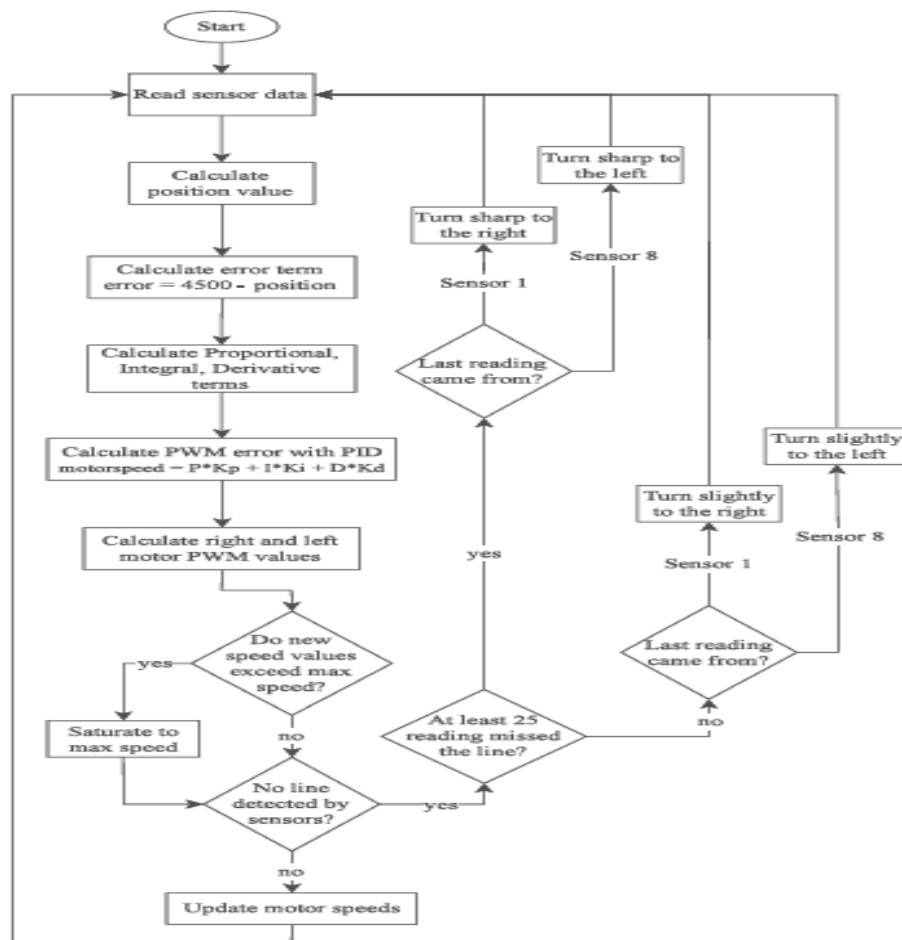


Figure 2 Flow Chart

Code:

```
#include <NewPing.h>
#include <Servo.h>

// Pin configuration for L298N
#define L298N_IN1 3
#define L298N_IN2 4
#define L298N_IN3 5
#define L298N_IN4 6
#define L298N_ENA 11 // PWM pin for Motor A enable
#define L298N_ENB 10 // PWM pin for Motor B enable

// Pin configuration for ultrasonic sensor
#define TRIGGER_PIN 7
#define ECHO_PIN 8

// Pin configuration for IR sensors
#define LEFT_IR_PIN 12
#define RIGHT_IR_PIN 13
#define th 880
// QTR sensor configuration
const uint8_t SensorCount = 6;
uint16_t sensorValues[SensorCount];

// Ultrasonic sensor configuration
#define MAX_DISTANCE 200 // Maximum distance to check for obstacles
NewPing sonar(TRIGGER_PIN, ECHO_PIN, MAX_DISTANCE);

// Motor speed value (0-255)
#define MOTOR_SPEED 55

// PID constants for line following
float Kp = 0.5;
float Ki = 0.0;
float Kd = 0.0;

// PID variables
float error = 0, previous_error = 0;
float P = 0, I = 0, D = 0;
float PID_value = 0;

// Motor speed constants
int base_speed = 55;
int max_speed = 70;
int ir_thr = 870;

void setup() {
```

```

Serial.begin(9600);
pinMode(L298N_IN1, OUTPUT);
pinMode(L298N_IN2, OUTPUT);
pinMode(L298N_IN3, OUTPUT);
pinMode(L298N_IN4, OUTPUT);
pinMode(L298N_ENA, OUTPUT);
pinMode(L298N_ENB, OUTPUT);
pinMode(LEFT_IR_PIN, INPUT);
pinMode(RIGHT_IR_PIN, INPUT);
pinMode(TRIGGER_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);
for (int i = 0; i <= SensorCount; i++) {
  Serial.print("Sensor "); Serial.print(i); Serial.print(": ");
  Serial.println(analogRead(A0 + i));
  delay(500);
}

}

void loop() {
  // Read sensor values from QTR sensors
  for (uint8_t i = 0; i < SensorCount; i++) {
    sensorValues[i] = analogRead(A0 + i);
  }

  // Read IR sensors
  bool leftIR = digitalRead(LEFT_IR_PIN);
  bool rightIR = digitalRead(RIGHT_IR_PIN);

  // Read ultrasonic distance
  delay(50);
  int distance = sonar.ping_cm();

  // Obstacle detection
  if (distance > 0 && distance < 15) {
    stopMotors();
    delay(500);
    // Reverse and turn
    moveBackward();
    delay(500);
    stopMotors();
    delay(500);
    turnLeft(); // or turnRight() as per your preference
    delay(500);
    stopMotors();
    delay(500);
  } else {
    // Line following logic

```

```

        lineFollow();
    }
}

void moveForward() {
    digitalWrite(L298N_IN1, HIGH);
    digitalWrite(L298N_IN2, LOW);
    digitalWrite(L298N_IN3, HIGH);
    digitalWrite(L298N_IN4, LOW);
    analogWrite(L298N_ENA, MOTOR_SPEED);
    analogWrite(L298N_ENB, MOTOR_SPEED);
}

void moveBackward() {
    digitalWrite(L298N_IN1, LOW);
    digitalWrite(L298N_IN2, HIGH);
    digitalWrite(L298N_IN3, LOW);
    digitalWrite(L298N_IN4, HIGH);
    analogWrite(L298N_ENA, MOTOR_SPEED);
    analogWrite(L298N_ENB, MOTOR_SPEED);
}

void turnLeft() {
    digitalWrite(L298N_IN1, LOW);
    digitalWrite(L298N_IN2, HIGH);
    digitalWrite(L298N_IN3, HIGH);
    digitalWrite(L298N_IN4, LOW);
    analogWrite(L298N_ENA, MOTOR_SPEED);
    analogWrite(L298N_ENB, MOTOR_SPEED);
}

void turnRight() {
    digitalWrite(L298N_IN1, HIGH);
    digitalWrite(L298N_IN2, LOW);
    digitalWrite(L298N_IN3, LOW);
    digitalWrite(L298N_IN4, HIGH);
    analogWrite(L298N_ENA, MOTOR_SPEED);
    analogWrite(L298N_ENB, MOTOR_SPEED);
}

void stopMotors() {
    digitalWrite(L298N_IN1, LOW);
    digitalWrite(L298N_IN2, LOW);
    digitalWrite(L298N_IN3, LOW);
    digitalWrite(L298N_IN4, LOW);
    analogWrite(L298N_ENA, 0);
    analogWrite(L298N_ENB, 0);
}

```

```

void lineFollow() {
    // Read QTR sensor values
    int leftmost_sensor_value = sensorValues[0];
    int left_sensor_value = sensorValues[1];
    int center_left_sensor_value = sensorValues[2];
    int center_sensor_value = sensorValues[3];
    int center_right_sensor_value = sensorValues[4];
    int right_sensor_value = sensorValues[5];
    int rightmost_sensor_value = sensorValues[6];

    // Check if all sensors are on the line
    if (leftmost_sensor_value < th && left_sensor_value < th &&
center_left_sensor_value < th && center_sensor_value < th &&
center_right_sensor_value < th && right_sensor_value < th) {
        stopMotors();
    } else {
        // Calculate the error
        error = (leftmost_sensor_value * -0.3) + (left_sensor_value * -0.2) +
            (center_left_sensor_value * -0.1) +
            (center_sensor_value * 0.1) + (center_right_sensor_value *
0.2) +
            (right_sensor_value * 0.3);

        // Proportional term
        P = error;

        // Integral term
        I += error;

        // Derivative term
        D = error - previous_error;

        // Calculate the PID value
        PID_value = (Kp * P) + (Ki * I) + (Kd * D);

        // Compute the motor speeds
        int left_motor_speed = base_speed - PID_value;
        int right_motor_speed = base_speed + PID_value;

        // Constrain the motor speeds to the maximum speed
        left_motor_speed = constrain(left_motor_speed, 0, max_speed);
        right_motor_speed = constrain(right_motor_speed, 0, max_speed);

        // Set the motor speeds and directions
        setMotorSpeedAndDirection(left_motor_speed, right_motor_speed);

        // Save the current error as the previous error for the next loop
    }
}

```

```

        previous_error = error;

        // Debugging output
        Serial.print("Error: ");
        Serial.print(error);
        Serial.print(" | PID Value: ");
        Serial.print(PID_value);
        Serial.print(" | Left Motor Speed: ");
        Serial.print(left_motor_speed);
        Serial.print(" | Right Motor Speed: ");
        Serial.println(right_motor_speed);
    }
}

void setMotorSpeedAndDirection(int left_speed, int right_speed) {
    // Set direction and speed for left motor
    if (left_speed >= 0) {
        digitalWrite(L298N_IN1, HIGH);
        digitalWrite(L298N_IN2, LOW);
    } else {
        digitalWrite(L298N_IN1, LOW);
        digitalWrite(L298N_IN2, HIGH);
        left_speed = -left_speed; // Make speed positive for PWM
    }
    left_speed = map(left_speed, 0, 100, 0, 255);
    analogWrite(L298N_ENA, left_speed);

    // Set direction and speed for right motor
    if (right_speed >= 0) {
        digitalWrite(L298N_IN3, HIGH);
        digitalWrite(L298N_IN4, LOW);
    } else {
        digitalWrite(L298N_IN3, LOW);
        digitalWrite(L298N_IN4, HIGH);
        right_speed = -right_speed; // Make speed positive for PWM
    }
    right_speed = map(right_speed, 0, 100, 0, 255);
    analogWrite(L298N_ENB, right_speed);
}

```

Hardware Implementation:

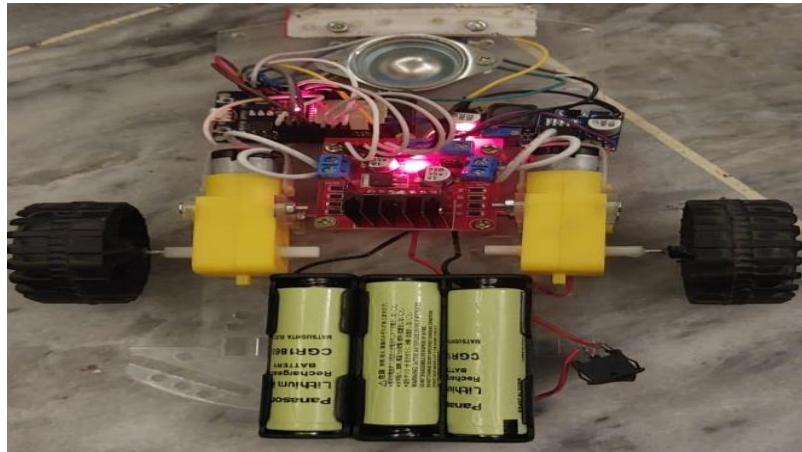


Figure 3 Hardware Setup

3. Experimental Setup & Results

Calibration:

- Sensors calibrated manually.
- Verified consistent black/white value separation via Serial Monitor.
- Fine-tuned detection threshold (under 300).

Performance Highlights:

- **PID Constants:** Tuned to $K_p = 0.8$, $K_i = 0.01$, $K_d = 2.0$.
- **Dynamic Adjustment:** Smooth tracking on curves and quick error correction.
- **Fallback Logic:** Robot halts within 500 ms if the line is lost.

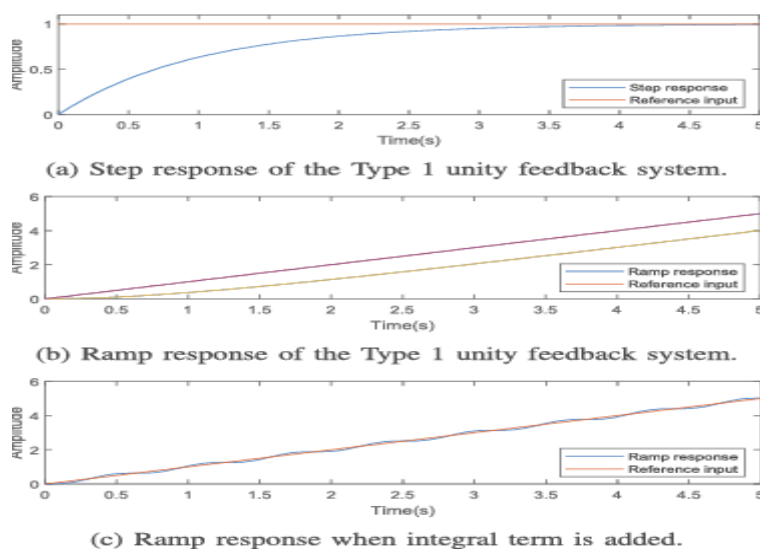


Figure 4 Time Domain Response

Testing Outcome:

- Successful line tracking on a black line track.
- Smooth handling of straight and curved paths.
- Reliable stop behavior on line loss.

4. Discussion, Challenges & Future Work

Challenges:

- Sensor noise due to lighting and surface conditions.
- PID tuning needed iteration for optimal performance.
- L298N driver had voltage drop and heat concerns.

Potential Improvements:

- Adaptive PID for auto-tuning.
- Sensor fusion (e.g., IMU, ultrasonic) for robustness.
- Efficient motor drivers like TB6612FNG.
- Data logging and IoT integration for remote monitoring.

5. Conclusion

The project successfully delivered a **cost-effective, PID-controlled** line follower robot using **basic components** and **open-source libraries**. It demonstrated reliable performance, accurate tracking, and safe fallback behavior—validating its potential as a low-cost educational and research platform.

Future directions include advanced control algorithms (fuzzy logic, AI), expanded sensor setups, and IoT connectivity for real-time diagnostics and control.